

Unit 2: Data Link Layer, Error Control & Multiple Access

Handwritten Student Notebook

Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED Data Link Layer Framing: Bit stuffing (01111110 delimiter) and byte stuffing (ESC flags), physical MAC addressing. Address Resolution Protocol (ARP): ARP request/reply, packet format, ARP cache operation, ARP spoofing attacks. Er...

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Data Link Layer Duties & Framing Methods

§ 1

The **Data Link Layer (DLL)** transforms a raw, error-prone transmission facility into a reliable, hop-to-hop link for the Network Layer. It accomplishes this by packaging network-layer packets into discrete units called **frames**, regulating transmission speed via flow control, and detecting physical errors.

CORE RESPONSIBILITIES OF THE DATA LINK LAYER:

- **Framing:** Divides the stream of bits received from the physical layer into manageable data units (frames) with distinct header and trailer delimiters.
- **Physical Addressing:** Appends a header containing the physical hardware addresses (48-bit MAC addresses) of the sender and immediate receiver.
- **Flow Control:** Prevents a fast sender from swamping a slow receiver using feedback-based sliding window mechanisms.
- **Error Control:** Detects and retransmits damaged or lost frames by adding trailers containing Cyclic Redundancy Check (CRC) codes.
- **Access Control:** In broadcast channels, determines which device has permission to transmit at any given time (MAC sublayer).

Framing Techniques: Character Count vs Stuffing

Framing Technique	Delimiter / Mechanism	Operation & Vulnerability
Character Count	Integer in the frame header specifies total frame length.	Fatal flaw: If the count field is corrupted by a single bit flip, the receiver loses frame synchronization for all subsequent frames. Deprecated in modern networks.
Byte Stuffing (Byte-Oriented)	Special flag bytes (e.g. `FLAG` = 0x7E) delimit start and end of frames.	If the flag pattern occurs naturally within data, the sender inserts an escape byte (ESC = 0x7D) before it. If an `ESC` occurs in data, it is escaped with another `ESC` (ESC ESC). Receiver strips the stuffing `ESC`.

Framing Technique	Delimiter / Mechanism	Operation & Vulnerability
Bit Stuffing (Bit-Oriented)	Special bit pattern '01111110' (six consecutive 1s) delimits frame boundaries.	Rule: Whenever the sender encounters five consecutive 1s (11111) in user data, it automatically stuffs a '0' after the fifth 1. Receiver detects five 1s and discards the stuffed 0.

WORKED EXAMPLE: BIT STUFFING & UNSTUFFING:

Input Bit Stream: 01001111101111110101011111110

Step 1: Sender Stuffing (Insert 0 after every five consecutive 1s):

- 0100 11111 $\rightarrow$ stuff 0 $\rightarrow$ 0100 11111[0] 0
- 111111 $\rightarrow$ after five 1s, stuff 0 $\rightarrow$ 11111[0] 10
- 010 11111110 $\rightarrow$ after five 1s, stuff 0 $\rightarrow$ 010 11111[0] 110

Stuffed Stream Transmitted: 0100111110011111010010111110110

Step 2: Receiver Unstuffing: Receiver encounters five 1s followed by a 0, removes the 0, perfectly restoring the original data stream.

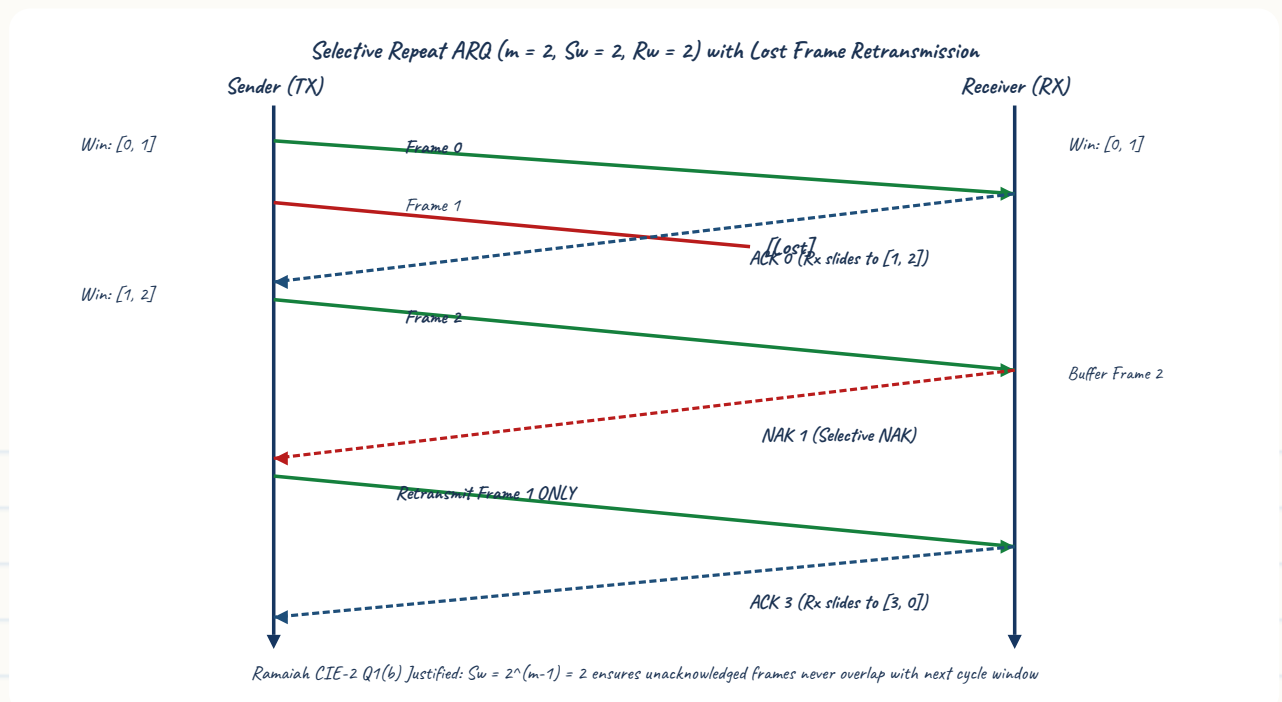


Figure 2.2: Selective Repeat ARQ Flow ($m = 2$, Sender Window $S_w = 2^{(m-1)} = 2$)

Ramaiah CIE-2 Q1(b) Justified: When frame 1 is lost, receiver buffers frame 2 and issues NAK 1. Only frame 1 is retransmitted.

QUICK RECALL

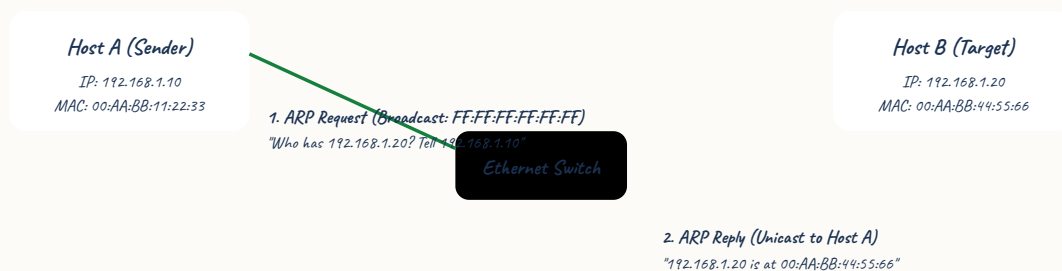
- DLL handles hop-to-hop delivery using 48-bit physical MAC addresses.
- Byte stuffing escapes special flag bytes with `ESC`.
- Bit stuffing inserts a `0` after every **five consecutive 1s** (11111) to prevent imitation of flag `01111110`.

Address Resolution Protocol (ARP)

§ 2

While the Network Layer routes packets based on 32-bit logical IPv4 addresses, physical local area networks (such as Ethernet) only deliver frames using 48-bit physical MAC addresses. The **Address Resolution Protocol (ARP)** dynamically maps a known IP address to its unknown physical MAC address on the local link.

ARP Protocol Exchange: Broadcast Request & Unicast Reply



ARP Packet Fields (28-Byte Payload)

Field Name	Length	Description
Hardware Type (HTYPE)	16 bits	Physical network protocol (Ethernet = `1`).
Protocol Type (PTYPE)	16 bits	Network layer protocol (IPv4 = `0x0800`).
Hardware Length (HLEN)	8 bits	Length of hardware address in bytes (Ethernet = `6`).
Protocol Length (PLEN)	8 bits	Length of logical address in bytes (IPv4 = `4`).

Field Name	Length	Description
Operation Code (OPER)	16 bits	`1` for ARP Request, `2` for ARP Reply.
Sender Hardware / IP	6 + 4 bytes	MAC address and IP address of the transmitting node.
Target Hardware / IP	6 + 4 bytes	Target MAC (00:00:00:00:00:00 in request) and Target IP.

SECURITY WARNING: ARP SPOOFING / POISONING:

Because ARP is completely stateless and does not authenticate replies, an attacker on the same local subnet can send unsolicited (gratuitous) fake ARP replies claiming their MAC address belongs to the default gateway's IP. All outbound traffic from victim hosts is then silently routed through the attacker (Man-in-the-Middle attack).

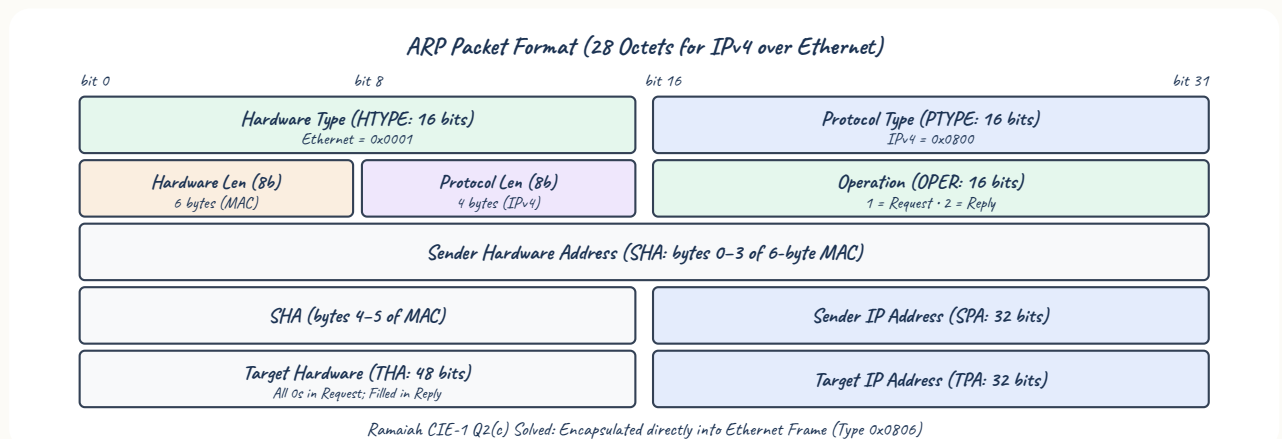


Figure 2.1: 28-Byte ARP Packet Structure over Ethernet & IPv4 (RFC 826)

As required in Ramaiah CIE-1 Exam: Maps 32-bit IP addresses to 48-bit MAC addresses within local broadcast domains.

QUICK RECALL

- ARP maps 32-bit IPv4 address → 48-bit MAC address.
- ARP Request is a **Broadcast** (FF:FF:FF:FF:FF:FF); ARP Reply is a **Unicast**.
- ARP Cache stores mappings temporarily with a Time-to-Live (TTL).

Error Detection, Correction & Block Coding

Electromagnetic noise and interference inevitably introduce errors during physical transmission. Errors are classified as **single-bit errors** (only 1 bit inverted in the entire data unit) or **burst errors** (2 or more consecutive bits corrupted, where the burst length is measured from the first to the last corrupted bit).

Hamming Distance

HAMMING DISTANCE DEFINITION:

The **Hamming distance** $d(x, y)$ between two binary words of equal length is the number of bit positions in which the corresponding bits differ. It is easily calculated by taking the bitwise exclusive-OR ($x \oplus y$) and counting the number of 1s (Hamming weight).

Example: $x = 10101$ and $y = 11110$. $x \oplus y = 01011$ (three 1s)
 $\implies d(x, y) = 3$.

The **minimum Hamming distance** $d_{\min}$ of a code is the smallest Hamming distance between all distinct pairs of valid codewords in that code.

Mathematical Criteria for Detection and Correction

HAMMING DISTANCE CRITERIA:

1. To detect up to s errors:

$$d_{\min} \geq s + 1$$

Intuition: If a transmitted codeword incurs up to s bit flips, the received pattern will lie at a distance $\leq s$ from the original codeword, but at least 1 flip away from any other valid codeword, guaranteeing detection.

2. To correct up to t errors:

$$d_{\min} \geq 2t + 1$$

Intuition: Spheres of radius t drawn around valid codewords must never intersect. A received word with up to t errors is uniquely closer to the original transmitted codeword than to any other.

QUICK RECALL

- Hamming distance $d(x, y)$ = number of differing bits = weight of $x \oplus y$.
- Error Detection: $d_{\min} \geq s + 1$. Example: to detect 2 errors, $d_{\min} \geq 3$.
- Error Correction: $d_{\min} \geq 2t + 1$. Example: to correct 1 error, $d_{\min} \geq 3$.

Cyclic Redundancy Check (CRC)

§ 4

Cyclic Redundancy Check (CRC) is an exceptionally powerful, hardware-friendly error-detecting code used across Ethernet, Wi-Fi, and storage systems. It treats binary bit streams as algebraic polynomials with binary coefficients and performs **modulo-2 polynomial division**.

CRC_SENDER_AND_RECEIVER_ALGORITHM:

Let dataword D have k bits, and generator polynomial $G(x)$ have degree r ($r + 1$ bits):

1. **Sender Step 1:** Append r zeros to the right of dataword D , forming $D \cdot 2^r$.
2. **Sender Step 2:** Divide the appended bit stream $D \cdot 2^r$ by the divisor $G(x)$ using **modulo-2 binary division** (bitwise XOR subtraction, no carries or borrows).
3. **Sender Step 3:** The division yields a quotient (discarded) and an r -bit remainder R , called the **CRC**.
4. **Sender Step 4:** Append the r -bit CRC remainder to original dataword D . Transmit codeword $T = D \cdot 2^r \oplus R$.
5. **Receiver Step:** Divide the received codeword T' by the identical generator $G(x)$. If the remainder is **strictly all zeros**, accept the frame; otherwise, an error occurred — discard frame!

WORKED NUMERICAL · FULLY VERIFIED PASS [10 MARKS · EXAM FAVOURITE]

CRC Modulo-2 Division at Sender and Receiver

Given Dataword $D = 100100$ and Generator polynomial $G(x) = x^3 + x^2 + 1$.

1. Determine the binary representation and degree of $G(x)$.
2. Compute the transmitted codeword using modulo-2 binary division.
3. Verify error detection at the receiver when the transmission is error-free.

Step 1: Generator Polynomial Properties:

$G(x) = 1 \cdot x^3 + 1 \cdot x^2 + 0 \cdot x^1 + 1 \cdot x^0 \implies \mathbf{1101}$. Degree $r = 3$.

Step 2: Append $r = 3$ zeros to Dataword:

Appended dividend = 100100000 (9 bits).

Step 3: Modulo-2 Long Division at Sender:

```

1101 ) 100100000 ( Quotient: 110010
      1101
      ----
        1000
        1101
        ----
          1010
          0000
          ----
            1010
            1101
            ----
              1110
              1101
              ----
                0110
                0000
                ----
                  001 ← Remainder (CRC)
  
```

The 3-bit remainder is $R = \mathbf{001}$.

Step 4: Transmitted Codeword:

Codeword $T = 100100$ concatenated with $001 = \mathbf{100100001}$

Step 5: Receiver Verification:

Receiver divides received codeword 100100001 by divisor 1101:

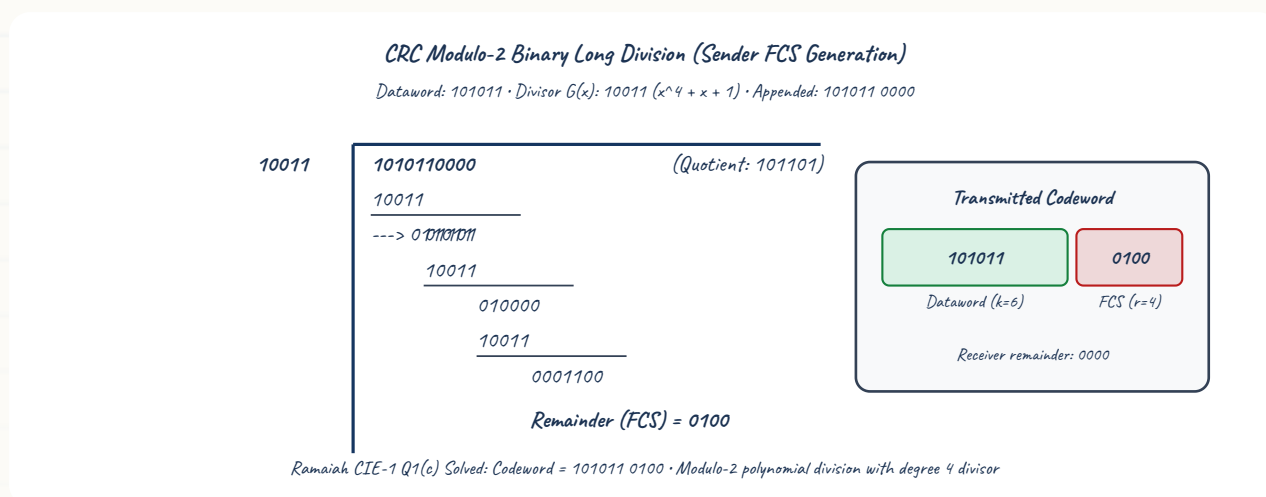


Figure 2.3: CRC Modulo-2 Polynomial Long Division (Ramaiah CIE-1 Q1.c Solved)

Dataword 101011 · Divisor 10011 ($x^4 + x + 1$) · Remainder (FCS) 0100 · Transmitted Codeword: 101011 0100.

[Interactive Visualizer Online]: [Interactive CRC-4 Modulo-2 Calculator & Error Injector](#). Explore the step-by-step visualizer on the live course notes page.

QUICK RECALL

- Append $r = \deg(G(x))$ zeros to dataword before dividing.
- Modulo-2 division uses XOR (same bits = 0, different bits = 1; NO carries/borrows).
- Codeword = Dataword + Remainder. Receiver gets remainder 0 if error-free.

The Internet Checksum Algorithm

§ 5

The **Internet Checksum** is utilized by IP, TCP, and UDP. It treats data as a sequence of 16-bit integers and computes their sum using **one's complement arithmetic** with end-around carry.

CHECKSUM COMPUTATION STEPS:**At Sender:**

1. Divide data into 16-bit words. (If length is odd, pad with a trailing zero byte).
2. Sum all 16-bit words together.
3. If a carry bit overflows beyond 16 bits (bit 16), wrap the carry around and add it to the least significant bit (LSB). Repeat until no carry remains.
4. Take the bitwise NOT (one's complement) of the 16-bit sum. This is the **Checksum**.
5. Send the data words along with the checksum.

At Receiver: Sum all received 16-bit words including the checksum using one's complement. If the result is all 1s ($0xFFFF$), which when complemented yields $0x0000$, the packet is accepted!

WORKED NUMERICAL · FULLY VERIFIED PASS [8 MARKS]

Compute 16-bit Internet Checksum for Given Words

Calculate the Internet Checksum for the following three 16-bit hexadecimal words:

Word 1: $0x4500$, Word 2: $0x003C$, Word 3: $0x1C46$

Step 1: Standard Binary/Hexadecimal Addition:

$$\begin{array}{r}
 0x4500 \\
 +0x003C = 0x453C \\
 +0x1C46 = 0x6182
 \end{array}$$

Since the sum $0x6182$ fits in 16 bits without carry overflow, wrapped sum = $0x6182$.

Step 2: Take One's Complement (Bitwise Inversion):

$$\text{Sum} = 0110\ 0001\ 1000\ 0010_2\ (0x6182)$$

$$\text{Checksum} = \sim \text{Sum} = 1001\ 1110\ 0111\ 1101_2 = 0x9E7D$$

Step 3: Receiver Verification:

$0x4500 + 0x003C + 0x1C46 + 0x9E7D = 0x6182 + 0x9E7D = 0xFFFF$
 Inverting $0xFFFF$ gives $0x0000$. The data is completely verified!

QUICK RECALL

- Internet Checksum uses 16-bit one's complement addition with end-around carry.
- Sender checksum = bitwise NOT of sum.
- Receiver sum of all words + checksum must equal all 1s (0xFFFF).

Random Access Protocols: ALOHA & CSMA/CD

§ 6

In a broadcast network, multiple stations share a single communication channel.

Random Access (Contention) Protocols allow stations to transmit whenever they have data, resolving collisions when simultaneous transmissions overlap.

1. Pure ALOHA vs Slotted ALOHA Throughput Analysis

Let G be the offered load (average number of frame generation requests per frame transmission time T_{fr}):

Protocol	Vulnerable Window	Throughput Formula	Max Load (G^*)	Max Throughput (S_{max})
Pure ALOHA	$2 \times T_{fr}$	$S = Ge^{-2G}$	$G = 0.5$	$\frac{1}{2e} \approx 18.4\%$
Slotted ALOHA	T_{fr}	$S = Ge^{-G}$	$G = 1.0$	$\frac{1}{e} \approx 36.8\%$

2. Carrier Sense Multiple Access (CSMA)

Before transmitting, stations "listen before speaking" (sense the carrier) to avoid obvious collisions:

- **1-Persistent CSMA:** If channel is idle, transmit immediately. If busy, continue sensing continuously until idle, then transmit with probability $p = 1$. (High collision probability when two waiting stations transmit simultaneously upon channel becoming idle).
- **Non-Persistent CSMA:** If idle, transmit immediately. If busy, wait a random backoff time before sensing again. (Reduces collisions, but introduces channel idle time).
- **p-Persistent CSMA:** Used in slotted channels. If idle, transmit with probability p ; defer to next slot with probability $1 - p$.

3. CSMA with Collision Detection (CSMA/CD)

Stations "listen while talking". If two signals collide on the wire, the collision energy spike is detected immediately, transmission is aborted, a 32-bit **jamming signal** is broadcast to alert all nodes, and stations invoke the **Binary Exponential Backoff** algorithm.

CSMA/CD MINIMUM FRAME LENGTH RULE:

For a station to reliably detect a collision before it finishes transmitting its frame, the frame transmission time T_{fr} must be at least twice the maximum end-to-end propagation delay t_{prop} (round-trip time):

$$T_{fr} \geq 2 \times t_{prop}$$

Since $T_{fr} = \frac{L_{min}}{\text{Bandwidth}}$, we have the fundamental dimensioning formula:

$$L_{min} \geq 2 \times t_{prop} \times \text{Bandwidth}$$

WORKED NUMERICAL · VERIFIED SCRIPT PASS [8 MARKS · CORE EXAM]**CSMA/CD Minimum Frame Size Calculation**

A network operating CSMA/CD has a transmission bandwidth of $R = 10$ Mbps, a maximum cable length of $d = 2500$ m, and a signal propagation velocity of $v = 2 \times 10^8$ m/s. Calculate the minimum frame size required for collision detection.

Step 1: Calculate One-Way Propagation Delay (t_{prop}):

$$t_{prop} = \frac{d}{v} = \frac{2500 \text{ m}}{2 \times 10^8 \text{ m/s}} = 1.25 \times 10^{-5} \text{ s} = 12.5 \mu\text{s}$$

Step 2: Calculate Round-Trip Time ($RTT = 2t_{prop}$):

$$RTT = 2 \times t_{prop} = 2 \times 12.5 \mu\text{s} = 25.0 \mu\text{s}$$

Step 3: Apply Minimum Frame Size Equation:

$$L_{min} = 2 \times t_{prop} \times R = (25.0 \times 10^{-6} \text{ s}) \times (10 \times 10^6 \text{ bps}) = \mathbf{250 \text{ b}}$$

$$L_{min} = \frac{250}{8} = \mathbf{31.25 \text{ bytes} \approx 32 \text{ bytes}}$$

Standard Ethernet Note: In IEEE 802.3 10BASE5 standard Ethernet, accounting for 4 repeaters and worst-case delay, round-trip time is standardized to $51.2 \mu\text{s}$, producing the famous minimum frame size of **512 bits = 64 bytes**.

QUICK RECALL

- Pure ALOHA max throughput: 18.4% at $G = 0.5$. Slotted ALOHA max throughput: 36.8% at $G = 1.0$.
- CSMA: 1-persistent (greedy), non-persistent (random wait), p-persistent (probabilistic).
- CSMA/CD requires $T_{fr} \geq 2t_{prop} \implies L_{min} \geq 2t_{prop} \times \text{Bandwidth}$.

Controlled Access & Channelization Protocols**§ 7**

Unlike random access where stations compete and collide, **Controlled Access** ensures collision-free communication by having stations consult each other before transmitting, while **Channelization** divides channel capacity across frequency, time, or orthogonal code space.

1. Controlled Access Protocols

Protocol	Operational Mechanism	Pros & Cons
Reservation	Time is divided into intervals. In each interval, a reservation frame precedes data frames. Stations declare their intent in small mini-slots before sending.	Collision-free; overhead of reservation mini-slots.
Polling	One device acts as Primary (controller); others are Secondaries. Primary queries secondaries via Poll (to receive data) or Select (to send data).	Centralized control; single point of failure at primary; polling overhead latency.
Token Passing	Stations organized in a logical ring. A special bit pattern called the Token circulates. A station holding the token has exclusive right to transmit.	Deterministic, fair under heavy loads; token maintenance overhead (lost token, duplicate token).

2. Channelization Protocols (FDMA, TDMA, CDMA)

- **FDMA (Frequency-Division Multiple Access)**: Available channel bandwidth is partitioned into discrete frequency bands separated by guard bands. Each station transmits continuously on its assigned band.
- **TDMA (Time-Division Multiple Access)**: Stations share the entire frequency bandwidth but are allocated synchronized time slots within periodic frames.
- **CDMA (Code-Division Multiple Access)**: All stations transmit simultaneously over the entire frequency bandwidth! Signals are separated by assigning each station an **orthogonal chipping code (Walsh sequence)** where:

$$\mathbf{c}_i \cdot \mathbf{c}_j = 0 \quad (i \neq j), \quad \mathbf{c}_i \cdot \mathbf{c}_i = \mathbf{m}$$

QUICK RECALL

- Controlled access: Reservation (mini-slots), Polling (Primary/Secondary), Token Passing (Token Ring).
- FDMA separates by frequency bands; TDMA separates by synchronized time slots.
- CDMA allows simultaneous transmission over full bandwidth using orthogonal Walsh codes.

One-Page Rapid Revision Sheet

REVISION

Framing & Stuffing

- Bit stuffing: insert `0` after five consecutive 1s (11111). Delimiter: `01111110`.
- Byte stuffing: escape flag byte with `ESC`.
- ARP: maps IP (32-bit) → MAC (48-bit). Broadcast Request, Unicast Reply.

Hamming Distance & Codes

- Hamming distance $d(x, y) = \text{weight}(x \oplus y)$.
- Detect s errors: $d_{\min} \geq s + 1$.
- Correct t errors: $d_{\min} \geq 2t + 1$.
- Modulo-2 division uses XOR (no carries/borrows).

CRC & Checksum

- Append $r = \deg(G(x))$ zeros to dataword before dividing.
- Codeword = Dataword + CRC Remainder.
- Internet Checksum: 16-bit one's complement sum with end-around carry; complement at end.
- Receiver checks for `0xFFFF`.

MAC Protocols

- Pure ALOHA: $S = Ge^{-2G}$, $S_{\max} = 18.4\%$ at $G = 0.5$.
- Slotted ALOHA: $S = Ge^{-G}$, $S_{\max} = 36.8\%$ at $G = 1.0$.
- CSMA/CD: $L_{\min} \geq 2t_{\text{prop}} \times \text{Bandwidth}$.
- CDMA: orthogonal chipping sequences ($\mathbf{c}_i \cdot \mathbf{c}_j = 0$).

Important Examination Questions

? QUESTIONS

2-Mark Questions (Short Answer)

Practice Problem:

2M What is the purpose of bit stuffing in framing?

Solution:

Answer: Bit stuffing prevents user data patterns from being falsely interpreted as frame boundary delimiters. Whenever the sender detects five consecutive 1s (11111) in data, it stuffs a 0, preventing the flag sequence 0111110 from appearing in the payload.

Practice Problem:

2M State the condition on minimum Hamming distance to correct t errors.

Solution:

Answer: $d_{\min} \geq 2t + 1$. This guarantees that spheres of radius t centered around valid codewords never overlap, allowing unique nearest-neighbor correction.

Practice Problem:

2M Compare maximum throughput of Pure ALOHA and Slotted ALOHA.

Solution:

Answer: Pure ALOHA has a maximum throughput of 18.4% ($1/2e$) at an offered load of $G = 0.5$, whereas Slotted ALOHA doubles maximum throughput to 36.8% ($1/e$) at $G = 1.0$ by eliminating partial frame overlaps.

5-Mark Questions (Medium Length)

Practice Problem:

5M Explain the working of the Address Resolution Protocol (ARP).

Solution:

Answer: When a source host needs to send an IP datagram to a local target but only knows the target's IP address, it broadcasts an ARP Request packet (FF:FF:FF:FF:FF:FF) asking "Who has IP X? Tell IP Y". Every host on the subnet receives the broadcast, but only the machine with IP X replies with an ARP Reply packet sent unicast back to the source containing its 48-bit physical MAC address. The source updates its ARP cache with a TTL.

Practice Problem:

5M Explain why CSMA/CD imposes a restriction on minimum frame size.

Solution:

Answer: In CSMA/CD, a station must detect a collision before it finishes transmitting its frame so it knows whether retransmission is needed. In the worst case, a collision occurs just before the signal reaches the furthest station at time t_{prop} , and the collision signal takes another t_{prop} to propagate back. Therefore, the transmission duration must satisfy $T_{fr} \geq 2t_{prop}$, yielding $L_{min} \geq 2t_{prop} \times \text{Bandwidth}$.

10-Mark Questions (Long Answer / Comprehensive)

Practice Problem:

10M Explain Cyclic Redundancy Check (CRC) with an example, and show division at both sender and receiver.

Solution:

Answer Summary: Detail polynomial modulo-2 arithmetic rules, zero appending, XOR long division at sender yielding CRC remainder, codeword concatenation, and receiver verification with remainder zero as solved step-by-step in Section 4.

Solved Questions from Past CIE-1 & CIE-2 Papers

ARCHIVE

The following questions have been transcribed from the official Ramaiah CIE-1 and CIE-2 test papers (Term 2025) in notes/cn/practice/cn-cie-1-and-2.pdf, solved with complete intermediate steps:

Ramaiah Internal Assessment - I (2025) · Q 1(c) [7 Marks · Blooms L3, CO2]

"Define cyclic codes and linear block codes. Given the dataword 101011 and the divisor 10011, generate the codeword using CRC polynomial division."

- **Linear Block Code:** A block code where the XOR sum (modulo-2 addition) of any two valid codewords is also a valid codeword in the codebook.
- **Cyclic Code:** A special linear block code where any cyclic left or right shift of a valid codeword produces another valid codeword.

- **CRC Modulo-2 Division:**

Dataword $d = 101011$ ($k = 6$ bits). Divisor $G(x) = 10011$ ($r = 4$ bits degree).

Append $r = 4$ zeros to dataword: 1010110000.

Performing binary modulo-2 long division ([Figure 2.3](#)):

Quotient = 101101, Remainder (FCS) = **0100**.

Transmitted Codeword = Dataword + FCS = **1010110100**.

Ramaiah Internal Assessment - I (2025) · Q 2(c) [7 Marks · Blooms L2, CO2]

"With a neat diagram, discuss the different fields of ARP packet."

See [Figure 2.1](#) for the complete 28-byte ARP packet layout. Field descriptions:

1. **Hardware Type (HTYPE: 16 bits):** Physical network protocol (e.g. 0x0001 for Ethernet).
2. **Protocol Type (PTYPE: 16 bits):** Higher-layer network protocol (e.g. 0x0800 for IPv4).
3. **Hardware Address Length (HLEN: 8 bits):** Length of MAC address in bytes (6 for Ethernet).
4. **Protocol Address Length (PLEN: 8 bits):** Length of logical address in bytes (4 for IPv4).
5. **Operation (OPER: 16 bits):** 1 for ARP Request (broadcast), 2 for ARP Reply (unicast).
6. **Sender Hardware Address (SHA: 48 bits):** 6-byte MAC address of sender.
7. **Sender Protocol Address (SPA: 32 bits):** 4-byte IPv4 address of sender.
8. **Target Hardware Address (THA: 48 bits):** All zeros in ARP Request; populated with destination MAC in Reply.
9. **Target Protocol Address (TPA: 32 bits):** 4-byte IPv4 address being resolved.

Ramaiah Internal Assessment - I (2025) · Q 3(a) [4 Marks · Blooms L4, C02]

"Compare bit stuffing and byte stuffing with suitable examples."

- **Byte (Character) Stuffing:** Used in character-oriented protocols (PPP). Delimiters are 1-byte flags (e.g. FLAG = 0x7E). Whenever the payload contains a byte identical to FLAG or ESC, an Escape byte (ESC = 0x7D) is stuffed immediately before it. Receiver removes ESC and preserves data.
- **Bit Stuffing:** Used in bit-oriented protocols (HDLC). Delimiter flag is 01111110 (six consecutive 1s). Whenever the sender detects **five consecutive 1s** in the payload stream, it automatically inserts an extra 0 bit, regardless of the subsequent bit. Receiver removes the 0 following five 1s to restore original stream.

Ramaiah Internal Assessment - 1 (2025) · Q 3(b) [4 Marks · Blooms L3, C02]
 "Calculate the Internet check sum for the text 'ISE DEPARTMENT' at sender and receiver side (assume following hexadecimal ASCII values: A=41, D=44, E=45, I=49, M=4D, N=4E, P=50, R=52, S=53, T=54, space=32)."

Step 1: Group 14 characters into 7 16-bit hexadecimal words:

- Word 1 ('IS'): 0x4953
- Word 2 ('E '): 0x4532 (using exam's specified space value 32)
- Word 3 ('DE'): 0x4445
- Word 4 ('PA'): 0x5041
- Word 5 ('RT'): 0x5254
- Word 6 ('ME'): 0x4D45
- Word 7 ('NT'): 0x4E54

Step 2: Sum the 16-bit words:

$$\text{Sum} = 0x4953 + 0x4532 + 0x4445 + 0x5041 + 0x5254 + 0x4D45 + 0x4E54 = \mathbf{0x21}$$

Step 3: Add wraparound carry:

$$\text{Carry} = 0x2 \implies 0x10F8 + 0x2 = \mathbf{0x10FA}$$

Step 4: Take one's complement at sender:

$$\text{Checksum} = \sim (0x10FA) = \mathbf{0xEF05}$$

Receiver Verification: Add all 7 words + Checksum = $0x210F8 + 0xEF05 =$

$0x2FFFF \implies 0xFFFF + 0x2 = 0x0001 + \dots = 0xFFFF$. Inverting $0xFFFF$ yields **0x0000** (Error Free).

(Note: If space is taken as standard ASCII 0x20 instead of given hex 32, Sum = 0x210E6, Checksum = 0xEF17).

Ramaiah Internal Assessment - 11 (2025) · Q 1(b) [6 Marks · Blooms L4, C04]
 "Consider $m=2$, and with necessary diagrams justify the statement: 'Size of the sender window must be equal to $2^{(m-1)}$ in Selective Repeat ARQ Protocol'."

Theoretical Justification:

In Selective Repeat ARQ, sequence numbers range from 0 to $2^m - 1$. To prevent ambiguous duplicate acceptance, the sum of sender and receiver window sizes must never exceed sequence space: $S_w + R_w \leq 2^m$. Since Selective Repeat requires symmetric flow control ($S_w = R_w$), we have $2S_w \leq 2^m \implies S_w \leq 2^{m-1}$.

For $m = 2$:

- Sequence space = $2^2 = 4$ (0, 1, 2, 3). Maximum $S_w = 2^{2-1} = 2$.
- **Why S_w cannot be 3:** If $S_w = 3$, sender transmits frames 0, 1, 2. Receiver receives all three and slides window to $[3, 0, 1]$. If all three ACKs are lost, sender times out and retransmits frame 0. Receiver window $[3, 0, 1]$ includes 0, so it erroneously accepts duplicate frame 0 as NEW frame 0 of the next cycle!
- **With $S_w = 2$:** Sender transmits frames 0, 1. Receiver window slides to $[2, 3]$. If retransmission of frame 0 arrives, receiver immediately recognizes $0 \notin [2, 3]$ and drops the duplicate without error ([Figure 2.2](#)).

Verified Practice Problems with Hidden Answers

PRACTICE

Test your problem-solving speed on these 6 exam-pattern numericals. Click each card to reveal the complete step-by-step verified solution:

PRACTICE 1 · HAMMING DISTANCE BOUNDS [3 MARKS]

A block code has a minimum Hamming distance of $d_{\min} = 5$. How many bit errors can it guarantee to (a) detect, and (b) correct?

Practice Problem:

Show Step-by-Step Solution

Solution:

- Error detection: $d_{\min} \geq s + 1 \implies s = d_{\min} - 1 = 5 - 1 = 4$ bit errors.
- Error correction: $d_{\min} \geq 2t + 1 \implies 2t \leq 4 \implies t = \lfloor (5 - 1)/2 \rfloor = 2$ bit errors.

Result: Detects up to 4 errors · Corrects up to 2 errors

PRACTICE 2 · CRC POLYNOMIAL DIVISION [5 MARKS]

Given dataword 110101 and divisor polynomial $G(x) = x^3 + x + 1$ (1011), determine the transmitted codeword.

Practice Problem:

Show Step-by-Step Solution

Solution:

Degree of divisor $r = 3$. Append 3 zeros: 110101000.

Performing modulo-2 division by 1011:

- $1101 \oplus 1011 = 110 \rightarrow 1100$
- $1100 \oplus 1011 = 111 \rightarrow 1111$
- $1111 \oplus 1011 = 100 \rightarrow 1000$
- $1000 \oplus 1011 = 011 \rightarrow 0110$
- Remainder (FCS) = **110**.

Transmitted Codeword: 110101 110

PRACTICE 3 · INTERNET CHECKSUM [4 MARKS]

Compute the 16-bit Internet checksum for the two hexadecimal words $0x466F$ and $0x7275$.

Practice Problem:

Show Step-by-Step Solution

Solution:

$$0x466F + 0x7275 = 0xB8E4$$

No wraparound carry since $\text{sum} \leq 0xFFFF$.

$$\text{Checksum} = \sim (0xB8E4) = \mathbf{0x471B}$$

Result: Checksum = $0x471B$

PRACTICE 4 · PURE ALOHA THROUGHPUT [4 MARKS]

A pure ALOHA channel transmits 200-bit frames on a shared 50 kbps link. If the network generates 100 frames/second, what is the throughput S ?

Practice Problem:

Show Step-by-Step Solution

Solution:

Frame transmission time: $T_{fr} = 200/50000 = 0.004 \text{ s} = 4 \text{ ms}$.

Offered load: $G = N \times T_{fr} = 100 \times 0.004 = \mathbf{0.4}$.

$$S = Ge^{-2G} = 0.4 \times e^{-2(0.4)} = 0.4 \times e^{-0.8} \approx 0.4 \times 0.44933 = \mathbf{0.1797}$$

Throughput: $S = 17.97\%$ (Carrying 44.9 frames/sec successfully)

PRACTICE 5 · CSMA/CD MINIMUM FRAME SIZE [5 MARKS]

In a 1 Gbps Ethernet network operating over a distance of 1 km ($v = 2 \times 10^8$ m/s), what is the minimum frame size required for reliable collision detection?

Practice Problem:

Show Step-by-Step Solution

Solution:

$$T_{prop} = \frac{1000 \text{ m}}{2 \times 10^8 \text{ m/s}} = 5 \times 10^{-6} \text{ s} = 5 \mu\text{s}$$

$$\text{Round Trip Time (RTT)} = 2 \times T_{prop} = 10 \mu\text{s}$$

$$\text{Min Frame Bits} = \text{RTT} \times \text{Bandwidth} = (10 \times 10^{-6} \text{ s}) \times (10^9)$$

$$\text{Min Frame Bytes} = \frac{10,000}{8} = \mathbf{1250 \text{ Bytes}}$$

Result: Minimum Frame Size = 1250 Bytes (10,000 bits)

PRACTICE 6 · STOP-AND-WAIT EFFICIENCY [5 MARKS]

A 1 Mbps point-to-point link has a one-way propagation delay of 20 ms. If frames are 1000 Bytes and ACKs are negligible, calculate Stop-and-Wait ARQ efficiency.

Practice Problem:

Show Step-by-Step Solution

Solution:

$$T_{trans} = \frac{8000 \text{ bits}}{10^6 \text{ bps}} = 8 \text{ ms}$$

$$\text{Round Trip Delay} = 2 \times 20 \text{ ms} = 40 \text{ ms}$$

$$\text{Cycle time} = T_{trans} + 2T_{prop} = 8 + 40 = 48 \text{ ms}$$

$$\eta = \frac{T_{trans}}{T_{trans} + 2T_{prop}} = \frac{8}{48} = \frac{1}{6} \approx \mathbf{16.67\%}$$

Result: Efficiency $\eta = 16.67\%$